



1st International Conference on Soft Computing Application in Smart Grid and Renewable Energy (SAGAR2027)

26th & 27th February , 2027

Department of Electrical & Computer Engineering
&

Department of Electrical Engineering
Ajay Binay Institute Of Technology (Autonomous), Cuttack



About SAGAR2027 :

The **1st International Conference on Soft Computing Application in Smart Grid and Renewable Energy (SAGAR2027)** will be held on **February 26–27, 2027**, at **ABIT(Autonomous)** in **hybrid mode**. The conference is jointly organized by the **Department of Electrical and Computer Engineering** and **Department of Electrical Engineering**.

SAGAR2027 focuses on the application of **soft computing techniques** which includes **artificial intelligence, machine learning, neural networks, fuzzy logic, evolutionary algorithms, optimization, data science, and analytics** that address challenges in smart grids and renewable energy systems.

The conference aims to bring together researchers, academicians, industry professionals, and students to discuss recent advances in intelligent technologies for **power systems, renewable energy integration, energy management, forecasting, control, optimization, and reliability enhancement**. It provides a platform for knowledge sharing, collaboration, and exploration of future trends in sustainable and smart energy systems.

About Ajay Binay Institute of Technology (ABIT) Autonomous college :

Ajay Binay Institute of Technology (ABIT) is approved by **AICTE**, affiliated with **BPUP**, and has been granted **Autonomous Status by UGC**. ABIT group operates an **Engineering College** and an **Industrial Training Institute** in Cuttack, Odisha.

The group was established in **1993** with the aim to improve the availability of **skilled technical manpower** for industries as well as to enhance reach of technical education to rural Odisha.

The Institute is approved by **AICTE, New Delhi**, affiliated to **Biju Patnaik University of Technology, Rourkela** and **accredited by NAAC**. ABIT has a balanced distribution of technical streams with equal emphasis on the **core sector** and **IT sector**.

Important Date :

- **Submission Open:** July 6, 2026
- **Submission Deadline:** Nov 20, 2026
- **Acceptance Notification:** Dec 27, 2026
- **Author Registration:** Jan 1, 2027
- **Camera Ready Paper Submission:** Jan 15, 2027
- **Conference dates:** February 26-27, 2027

Registration fee

Category	Indian Author	Foreign Author
Full Registration	6000 INR	150 USD
Participation Fee	2000 INR	50 USD

*Note:

- **Registration is valid for UG/PG students and PhD scholar with valid Id card, Academician and Industry professionals.**
- **At least one author registration is mandatory for each accepted paper.**
- **Co-authors who wish to attend the conference in offline mode must register under the Attendee category.**

***Accepted and presented papers will be considered for publication in Springer Proceedings (subject to Springer approval). The proceedings are expected to be indexed in major databases such as Scopus.**

Venue :

Department of Electrical and Computer Engineering (ECE)
and Department of Electrical Engineering,
AJAY BINAY INSTITUTE OF TECHNOLOGY (AUTONOMOUS),
Markat Nagar, Cuttack Development Authority, Sector – 1,
Cuttack – 753014

SAGAR2027 Website



<https://sagar2027.vercel.app/>

For Paper Submission



Contact Details :

+91 9853536400, +91 7978600770, +91 7978905972
sagar.abit2027@abit.edu.in,
sagar.abit27@gmail.com

ADVISORY COMMITTEE :

Chief Patron

- Er. Payal Mohapatra, Director & President of the ABIT PMCA Society
- Er. Satyadarshi Mishra, Head-Governing Body, ABIT

Patron

- Prof. Leena Samantaray, Principal, ABIT

Honorary Chairs

- Prof. Ming Yang, Kennesaw State University
- Prof. Ravita Lamba, IIT Roorkee
- Prof. Debashis Chatterjee, Jadavpur University

PROGRAM COMMITTEE :

General Chairs

- Prof. Kalyan Chatterjee, Professor, ISM Dhanbad
- Prof. Subroto Banerjee, NIT Durgapur
- Prof. Sabyasachi Pattanaik, Dean (R&D, CS), ABIT Cuttack
- Prof. Dulu Pattanaik, Group Academic Head, ABIT Cuttack

Convenor

- Dr. Rupali Mohanty, ABIT Cuttack

Co-convenor

- Dr. Samarjeet Satapathy, ABIT Cuttack

Organising Chairs

- Dr. Ashok Kumar Mahapatra, ABIT Cuttack
- Er. Satya Ranjan Das, ABIT Cuttack
- Prof. Sujata Dash, Nagaland University
- Prof. Partha Pratim Sarangi, KIIT-DU
- Prof. Srikant Mohapatra, KIIT-DU

Conference Secretaries

- Er. Prasantini Samal, ABIT Cuttack
- Er. Soumya Sidhartha Ray, ABIT Cuttack
- Er. Chandan Mandal, ABIT Cuttack

Publication Chairs

- Prof. Sachi Nandan Mohanty, VIT Andhra Pradesh University
- Prof. Manas Ranjan Kabat, Institute of Management and Information Technology, Cuttack
- Prof. Satyabrata Das, Veer Surendra Sai University of Technology, Burla
- Prof. Narayan Nahak, SOA-DU, Odisha
- Prof. Subhranshu Sekhar Tripathy, KIIT-DU, Odisha
- Prof. Satyajit Mohanty, VIT University, Vellore
- Prof. Manoj Praharaj, ABIT Cuttack
- Er. Rajalaxmi Panda, ABIT Cuttack

Organising Committee Members

- Er. Lopamudra Prusty, ABIT Cuttack
- Er. Jyoti Ranjan Sahoo, ABIT Cuttack
- Er. Shakti Prasad Mohanty, ABIT Cuttack
- Er. Ilyas Ahmed, ABIT Cuttack
- Er. Subhadra Behera, ABIT Cuttack
- Er. Madhuchanda Bhanjadeo, ABIT Cuttack
- Er. Rajib Nayak, ABIT Cuttack
- Er. Tushar Ranjan Mahalik, ABIT Cuttack
- Er. Bulbul Mallick, ABIT Cuttack
- Mr. Sumit Kumar Mallick, ABIT Cuttack
- Mr. Ayeshkanta Mohapatra, ABIT Cuttack

1st International Conference on Soft Computing Application in Smart Grid and Renewable Energy (SAGAR2027)



26th & 27th February, 2027

Department of Electrical & Computer Engineering
&

Department of Electrical Engineering

Ajay Binay Institute Of Technology (Autonomous), Cuttack

TECHNICAL TRACKS:

Track 1: Power Systems and Smart Grids

- Power generation, transmission, and distribution
- Power system operation and control
- Smart grid technologies and applications
- Renewable energy integration and grid integration challenges
- Grid resilience and reliability enhancement
- Energy storage systems and management
- Demand-side management and demand response strategies
- Smart Grid Planning and Operation

Track 2: AI & ML application and Cyber security

- AI-Based Intrusion Detection Systems (IDS)
- Deep Learning Applications
- Phishing and Spam Detection Using Machine Learning
- Malware Classification with Deep Learning
- Network Anomaly Detection Using AI
- Adversarial Attacks and Defenses in Machine Learning

Track 3: Application of optimization techniques

- power system optimization
- Vehicle Routing and Logistics Optimization
- Hyperparameter Optimization in Machine Learning
- Smart Grid Energy Management Optimization
- Production and Inventory Scheduling Optimization
- Resource Allocation in Cloud Computing Using Optimization Algorithms

Track 4: Power Electronics and Intelligent Control Systems

- Intelligent control of power converters
- Advanced inverter technologies
- Control strategies for renewable energy systems
- FACTS devices and smart grid applications
- Adaptive and robust control techniques
- Embedded systems for smart energy applications
- Automation in power systems
- Electric Vehicle and Charging station

Track 5: Emerging Technologies and Future Trends

- Internet of Things (IoT) and communication techniques
- Quantum computing applications in energy systems
- Green hydrogen technologies
- AI-driven carbon neutrality solutions
- Internet of Energy (IoE)
- Smart energy markets and trading
- Digital transformation in power systems
- Future challenges in renewable energy and smart grids

Paper Submission Link: <https://cmt3.research.microsoft.com/SAGAR2027>

TECHNICAL COMMITTEE:

- Prof. Sanjay Misra, Covenant University, Ota, Nigeria
- Prof. Milos Stojmenovic, Singidunum University, Serbia
- Prof. Georgios Tsaramirsis, Higher Colleges of Technology, Abu Dhabi
- Prof. Sven Groppe, University of Lübeck, Germany
- Prof. Mohammad Yamin, Abdulaziz University, Jeddah, Saudi Arabia
- Prof. Bichitrananda Guru, Registrar, ABIT Cuttack
- Prof. Prashant Kumar Pany, Dean (Academics), ABIT Cuttack
- Prof. Bhupesh Kumar Dewangan, OP Jindal University, Raigarh, India
- Prof. Mukesh Misra, Massey University, New Zealand
- Prof. Neha Mohanty, Department of Computer Science & Engineering, New Jersey Institute of Technology, USA
- Prof. Jinghua Groppe, University of Lübeck, Germany
- Prof. Abdulrhman Alshareef, King Abdulaziz University
- Prof. Gourav Gupta, Massey University, New Zealand
- Prof. Maria Avilés, Gijon Hospital, Asturias, Spain
- Prof. Izzat Al-Darraj, University of Baghdad, Baghdad
- Prof. Lie Li, Kennesaw State University, Atlanta, USA
- Prof. Abdulsattar Abdullah Hamad, University of Samarra, Iraq
- Prof. Vicente Garcia Diaz, University Oviedo, Spain
- Prof. Oscar Castillo, Tijuana Institute Technology, Mexico
- Prof. J P Das, University of Alberta, Canada
- Prof. Rajkumar Singh Rathore, Cardiff Met University, London, UK
- Prof. Osamah Ibrahim Khalaf, Al-Nahrain University, Baghdad, Iraq
- Prof. Udit Mamodiya, Poornima University, India
- Prof. Ghaida Muttashar Abdulsahib, University of Technology, Iraq
- Prof. Sameer Algburi, Al-Kitab University, Iraq
- Prof. Habib Hamam, FOE, University of Moncton, Canada
- Prof. Chinmay Chakraborty, BITS, Ranchi
- Prof. Priya Ranjan, UPES, Bidholi Campus, Dehradun
- Prof. Muthu Kumar, UPES, Bidholi Campus, Dehradun
- Prof. Mamata Mittal, DSE University, India
- Prof. Yasha Hasija, Delhi Technological University
- Prof. Anjali Priyadarshini, SRM University, Delhi
- Prof. Partha Pakre, NIT Silchar
- Prof. Babita Majhi, G.G.V, Central University, Bilaspur
- Prof. Tripti Swarnkar, NIT, Raipur, India
- Prof. Bichitrananda Patra, SOA University, India
- Prof. Vishal Goyal, GLA University
- Prof. H.S. Behera, VSSUT, Burla, India
- Prof. Hridayanath Thatoi, MSCB University, India
- Prof. Brojo Kishore Mishra, NIST Berhampur
- Prof. Balaji Rajendran, VNRVJIT, Hyderabad
- Prof. Swarupananda Bisoi, MSCB University, India
- Prof. Arunima Neog, NIT Nagaland
- Prof. Huidrom Sampon Meaglem Singh, NIT Manipur
- Prof. Santosh Kumar Singh, Nagaland University, India
- Prof. Bendang Ao, Nagaland University, India
- Prof. Anagha Bhattacharya, NIT Mizoram
- Prof. Siv Kumar Mishra, Director, Center for UG & PG Studies
- Prof. Ajit Barisal, OUTF, Bhubaneswar
- Prof. Satyasundara Mahapatra, PSIT, Kanpur
- Prof. Anjali Mohapatra, IIIT Bhubaneswar
- Prof. Pravat Ranjan Pati, Graphic Era-DU Dehradun, Uttarakhand
- Prof. Prakash Kumar Sarangi, VCE, Hyderabad
- Prof. Premananda Sahu, School of Computer Engineering, LPU
- Prof. Subhendu Sahoo, VIT, Chennai
- Prof. Amaresh Gantayat, SOA-DU, Odisha
- Prof. Bhabasis Mohapatra, SOA-DU, Odisha
- Prof. Abhijeet Choudhury, SOA-DU, Odisha
- Prof. Anshuman Bhuyan, GIET, Bhubaneswar

Tourist Attractions:

- | | | |
|--------------------------------|--------------------------------|----------------------------|
| • Barabati Stadium | • Barabati Fort | • Chilika Lake |
| • Silvercity Boat Club | • Odisha State Maritime Museum | • Lingaraj Temple |
| • Shree Jagannath Temple, Puri | • Netaji Birth Place Museum | • Dhauli santi stupa |
| • Konark Sun Temple | • Cuttack Chandi Temple | • Khandagiri and Udayagiri |
| | | • Puri Sea Beach |



Venue :

Department of Electrical and Computer Engineering (ECE)
and Department of Electrical Engineering,
AJAY BINAY INSTITUTE OF TECHNOLOGY (AUTONOMOUS),
Markat Nagar, Cuttack Development Authority, Sector – 1,
Cuttack – 753014

SAGAR2027 Website



<https://sagar2027.vercel.app/>

For Paper Submission



Contact Details :

+91 9853536400, +91 7978600770, +91 7978905972
sagar.abit2027@abit.edu.in,
sagar.abit27@gmail.com